Forecasting How Busy a Venue Will Be, and Counting How Many People Are There

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Full paper (39 pages) and technical version: flockcorp.com/research

Research questions

1. Can a system that watches each venue's own recent live readings beat the popular-times chart at saying how busy one venue will be at one hour? **2.** Can a model small enough to run on a credit-card-sized computer count the people in a room from a 160 × 120 thermal camera, so that busyness can one day be checked against real people without recording anyone?

What I built

- **The forecast in Flock,** my app for planning nights out. A data pipeline collects hourly live busyness readings for about 1,750 venues in Philadelphia, the Lehigh Valley and Miami, with every reading's provenance recorded. The served number corrects each venue's typical-week curve with a trailing offset from its recent readings and uses the latest reading directly. A gradient-boosted model with 106 inputs (weather, events, holidays, neighbours, venue attributes) is trained and gated against production, and only ships if it wins.
- **Owl,** a 718,550-parameter neural network that counts people on a FLIR Lepton 3.5 thermal camera in 14 to 16 ms on one Raspberry Pi 5 core, trained on a physics-based generator of thermal scenes and six licensed real datasets. It runs inside **Flux**, a venue sensor I designed, where only numbers leave the device, never an image.

How it was tested

Every forecast weight was fitted on earlier days and scored on **119,939 live readings** from 9 September to 8 October 2026 that it never saw, with date-block bootstrap intervals, persistence baselines and accuracy measured at every horizon from one to 24 hours. To check the findings against real crowds, I repeated the analyses on **5.3 million hourly counts from 315 public counting sites in seven countries** (pedestrian counters, subway entries, a gym and a university library), under an analysis plan whose pass rules were fixed and hashed before any data were downloaded. Owl was scored on six real benchmark sets and on sealed sets that no version had seen, with its and a heat-threshold rule's counts stored before any frame was counted.

Key findings

Result	Flock	Baseline
Exact crowd level (of five), 30-day test, one hour ahead	54.9%	30.2% popular times
Mean absolute error, same readings	18.00	26.40 (31.8% higher)
Exact level when the venue was read an hour earlier	77.0%	
Exact level two to 24 hours ahead	33.2–33.6%	30.2%
Real counts: one-hour error, curve + latest departure (15 datasets)	6.41	9.58 curve alone
Owl, frames counted exactly, five of six benchmark sets	82–96%	9–77% heat rule
Owl, sealed set of 300 frames no version had seen	94.3%	40.7% heat rule

- **The freshest reading of a venue is worth more than any correction to its typical-week curve.** The served number's gain over popular times comes from using the latest live reading; on 13 October production switched to a simpler rule built on that finding.
- **Real crowds behave differently from commercial busyness scores.** On public counts the typical-week curve is close to calibrated (slope 0.92) and a departure from it lasts for hours; against its own live scores the commercial curve is poorly calibrated (slope 0.50) and departures fade within about an hour. Carrying the latest departure onto the curve cut one-hour error by 33% and was best in 14 of 15 real-count datasets; where the counts are people inside a place, the hour-old count beats the curve, as on the commercial scores.
- **Owl generalises across the cameras it was trained on and, in its stress tests, ignores the warm objects that fool a threshold** (hands, mugs, screens), and on crowded classrooms it misses by one person on average where its predecessor missed by almost five.

Limitations

- The forecast is scored against a commercial provider's relative 0 to 100 estimates, not headcounts; the large gains are one hour ahead.
- Owl has not yet been measured inside a working venue; on a camera type it was not trained on, the heat rule did better, and sparse classrooms remain its weakest case.
- The benchmark sets that report Owl's headline numbers also guided its development; the sealed sets are the clean tests.

What comes next

A pre-registered retrain in early November 2026, scored on 14 held-out days under a decision rule fixed before its data existed; a sealed test on Flux's own camera in real rooms of one to fifteen people; and a venue study that pairs hand counts with Owl and with the provider's live scores, to measure for the first time how commercial busyness scores track the people actually inside.

My role. I designed the research questions, built the data pipeline, the serving system and the sensor hardware, ran the experiments and wrote the software, using AI coding and writing assistants as tools throughout. Data: commercial busyness readings (provider not named), OpenWeather (ODbL), Ticketmaster, TheSportsDB; thermal datasets from TU Wien, IISER Bhopal, PUT, Aalborg University, ISSAI and the SenSys 2021 authors, used under their licences or with permission; public counts from 13 city, transit and university publishers, credited in the paper.